

Novelty Check — Frame-Dependent Antenna Delay & Inter-Anchor UWB Self-Calibration

Prior-art and novelty assessment — prepared for
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Executive Summary

This document is a prior-art and novelty assessment of the AutoPos UWB inter-anchor self-calibration work, prepared ahead of submission to *Measurement* / *IEEE TIM*. Five claimed contributions were each checked against the literature, with an active search for disconfirming prior art. The conclusion: the work has exactly one defensible novelty — Contribution 1 (antenna delay is a frame-dependent, gauge-coupled quantity, demonstrated by a four-way, Vicon-validated ablation) — and even this is only **partially novel**. The underlying identifiability mathematics is already established in the acoustic / TDOA self-calibration literature [2, 4] and in surveying error-propagation theory [1]; therefore only the empirical UWB demonstration and the operational thesis (“antenna-delay constants must not be transferred across coordinate frames”) are new. Contributions 2 (the integrated system), 3 (broadcast scheduling) and 5 (Monte-Carlo dropout / DOP-based redundancy) are established or standard practice, defensible only as validated engineering and characterization. Contribution 4 (burst-poll SS-TWR for motion error) is partially novel at best and carries a naming collision with the existing AltDS-TWR method [16]. Recommendation: reposition the paper around Contribution 1’s empirical and operational result, explicitly cede the identifiability theory to the acoustic/TDOA lineage, demote Contributions 2/3/5 to validated engineering, and rename Contribution 4. So positioned, the paper is a clean, citable measurement contribution publishable at *Measurement* or *IEEE TIM*.

Table 1: Summary of verdicts

Contribution	Verdict	Closest prior art
C1 — Delay-scale gauge coupling (frame-dependent antenna delay)	PARTIALLY NOVEL	[1, 2, 4]
C2 — Integrated self-cal + joint delay system	ESTABLISHED-NOT-NOVEL	[10, 11, 20]
C3 — Broadcast 8-anchor 10 Hz scheduling	STANDARD-PRACTICE	[13]
C4 — Burst-poll SS-TWR (motion error)	PARTIALLY NOVEL (naming risk)	[14, 15, 16]
C5 — Monte-Carlo keep-k / DOP redundancy	ESTABLISHED-NOT-NOVEL	[18, 19]

1 Contribution 1 — Delay–layout coupling / frame-dependent antenna delay

PARTIALLY NOVEL.

The claim is that, in range-only inter-anchor self-calibration, a constant additive antenna-delay term and the global geometry scale form a single coupled gauge degree of freedom. They are not separately identifiable from pairwise ranges alone, so an external metric constraint is required to fix the gauge; therefore antenna delay is frame-dependent, not a transferable hardware constant. The evidence is a four-way ablation: imposing externally-surveyed ground-truth anchor coordinates without re-estimating delays degrades tag localization (311 mm RMSE), while re-estimating delays in the imposed frame recovers and improves it (78 mm). The validation uses Vicon motion capture with SE(3)/Sim(3) alignment.

The closest prior art, ordered by disconfirming strength, is as follows. Lichti et al. [1] show that, in minimally constrained self-calibration, an unmodeled additive rangefinder offset produces near-linear scale-type patterns in range residuals, in a pulsed-radio/UWB context. This directly anticipates the additive-delay to scale coupling, although it does not frame it as a gauge freedom resolved by motion-capture ablation inside a tag-localization pipeline. Kuang and Åström [2] and Kuang et al. [3] are the most likely scooping lineage: TOA/TDOA-plus-unknown-offset is the mathematical isomorph of TWR-plus-antenna-delay, and those works characterize and resolve the affine/gauge ambiguities including the constant offset. Batstone et al. [4] estimate sender/receiver positions under an unknown constant offset, which is the structural analogue of constant antenna delay coupled to geometry. Crocco et al. [5] and Simayijiang et al. [6] give closed-form microphone-array self-calibration methods that handle unknown internal delays/offsets, showing that joint geometry+offset self-calibration is mature. Shah et al. [7] show that uncalibrated delay degrades localization even with correct positions, supporting the consequence, but they treat delay as a hardware constant, which is the assumption argued against. Shalaby et al. [8] operationally treat antenna delay as calibrate-once/transferable, making it the key contrast reference for the operational thesis.

The contribution still survives because none of the acoustic/surveying works close the loop with a UWB TWR hardware realization, a four-way ablation isolating delay re-estimation in an imposed frame, and Vicon SE(3)/Sim(3) ground truth. The novelty should be positioned as empirical and operational, not as new identifiability mathematics.

2 Contribution 2 — Integrated system

ESTABLISHED-NOT-NOVEL as a concept; defensible only as a validated open implementation.

The integrated system combines inter-anchor self-calibration, joint antenna-delay estimation, dual-layer anchor geometry, and custom broadcast SS-TWR firmware on DWM1001C/DW1000 while bypassing PANS/DRTLS. Piavanini et al. [9, 10] embed Gauss–Newton joint estimation of anchor geometry and antenna delays, which is the central pairing claimed here. Lutz et al. [20] jointly estimate anchor poses and UWB sensor-error-model parameters. Hamer and D’Andrea [11] built a self-calibrating UWB network with clock synchronization, per-module self-localization, and simultaneous multi-robot operation via TDOA, with no central coordination. Nguyen et al. [12] jointly estimate 3D anchor positions and range biases via continuous-time factor-graph calibration. Custom DW1000 broadcast firmware bypassing PANS/DRTLS is widespread.

The defensible angle is that the specific assembly — dual-layer geometry, broadcast SS-TWR custom firmware on DWM1001C, a gauge-aware workflow, and released Vicon-validated evidence — is a credible but incremental engineering contribution. It should be claimed as a validated, reproducible system, not as conceptual novelty.

3 Contribution 3 — Broadcast 8-anchor 10 Hz scheduling

STANDARD-PRACTICE. Confirmed.

The claimed scheduling result is 8999 fixes/300 s with a balanced 3-tag load, zero timeouts, and a runtime-reconfigurable layout. Broadcast/scalable operation is inherent to TDoA RTLS and to SnapLoc [13], which lets an unlimited number of tags self-localize at a theoretical upper bound of 2.3 kHz with decimeter accuracy on the off-the-shelf DW1000. One-to-many ranging is codified in IEEE 802.15.4z. The throughput/reliability numbers are a useful reliability and throughput characterization, not conceptual novelty.

4 Contribution 4 — “Alt SS-TWR burst-poll” for motion-error reduction ($\sim 8\times$)

PARTIALLY NOVEL; high naming-collision risk; needs sharper framing.

Sidorenko et al. [14] are the key reference to beat: they model constant-velocity motion as a linear pseudo-clock-drift and correct it, pre-empting the conceptual mechanism. Dotlic et al. [15] give CFO-corrected SS-TWR with fewer frames, overlapping the fast-SS-TWR-with-reduced-error goal. Shalaby et al. [17] reduce variance via signal-timing optimization, overlapping the timing-design angle. Neiryneck et al. [16] create the naming collision: “AltDS-TWR” already denotes a specific double-sided method whose error depends only on one device’s clock drift, and the term is now standard. The proposed single-sided burst-poll must be renamed in the manuscript.

A burst-poll SS-TWR specifically targeting motion-induced error with a quantified $\sim 8\times$ reduction is not exactly anticipated, but the mechanism is well-trodden. Novelty is marginal without a clean theoretical model distinguishing it from Sidorenko et al. [14].

5 Contribution 5 — Monte-Carlo keep-k dropout + stratified vertical-coverage + anchor-redundancy-to-accuracy guarantee

ESTABLISHED-NOT-NOVEL. Confirmed (DOP/CRLB territory).

Anchor-redundancy-to-accuracy and geometry-quality analysis are classical DOP/GDOP/WGDOP and CRLB results. Albaidhani and Alsudani [18] address anchor selection by GDOP for UWB, and Albaidhani et al. [19] address anchor selection via MSE/WGDOP. k-coverage is standard in sensor networks; the per-fix accuracy-guarantee framing mirrors GNSS integrity monitoring, RAIM, and protection levels; and the stratified vertical-coverage finding is the well-known z-axis GDOP effect. The Monte-Carlo keep-k study is a reasonable empirical robustness characterization, not a new method.

6 Conclusions

(a) Defensible single-sentence novelty statement.

“We provide the first explicit empirical demonstration, validated against motion-capture ground truth, that in range-only UWB inter-anchor self-calibration the additive antenna delay and the global geometric scale constitute a single non-identifiable gauge degree of freedom — so antenna delay must be re-estimated within whatever metric frame is imposed and is therefore not a transferable hardware constant.”

(b) Experiments that would strengthen each thin claim.

- C1: add an explicit observability / Fisher-information (rank-deficiency) analysis of the delay-scale Jacobian, exhibit the degenerate singular value / null-space direction numerically, cite the acoustic isomorphism [2, 4] and [1] directly, and replicate the 311→78 mm ablation across ≥ 2 independent anchor geometries.
- C2: head-to-head quantitative comparison against [10] and [11] on the same dataset; release firmware + dataset.
- C3: reframe as a reliability/latency benchmark vs [13] / a PANS-TDoA baseline.
- C4: derive a closed-form motion-error model for burst-poll SS-TWR and compare to [14] and [15] on identical motion profiles; rename the scheme.
- C5: replace/augment the Monte-Carlo study with a CRLB or GDOP bound and show the keep-k accuracy curve tracks the bound.

(c) Overall positioning. The work is publishable at *Measurement* or *IEEE TIM* if and only if repositioned around Contribution 1’s empirical and operational result, explicitly ceding the identifiability mathematics to the acoustic/TDOA self-calibration and surveying literature, demoting Contributions 2–3 and 5 to validated engineering/characterization, and strengthening or renaming Contribution 4. Claiming broad theoretical novelty for the gauge coupling would invite a strong reject from any reviewer familiar with the [2] / [4] lineage. The operational thesis (“do not transfer antenna-delay constants across coordinate frames / deployments”), backed by the ablation, is a clean, citable, useful result — that is where the paper’s value lives.

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